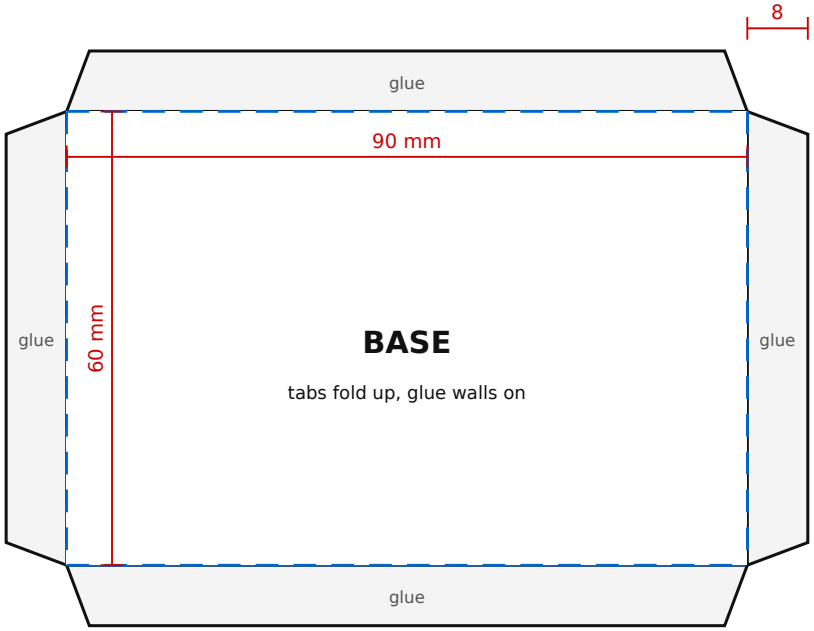
Robotics Project #13 · Templates 1/5 (1:1 scale)

Wheelbarrow Walker Robot

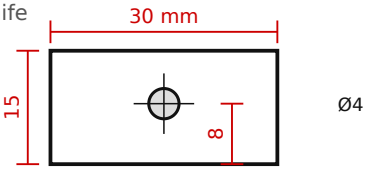
PRINT AT 100% / ACTUAL SIZE
Do NOT 'fit to page'
Check the 100 mm ruler first!

cut (solid) — — — fold (dashed) ◯ hole: grown-up pokes with a pencil/awl red = size in mm

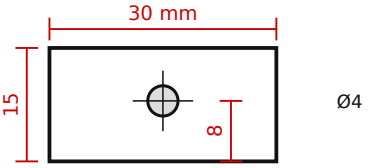
3 mm corrugated cardboard · grown-up cuts with craft knife



Barrow base 90x60 + 8 mm glue tabs (wheels roll freely)

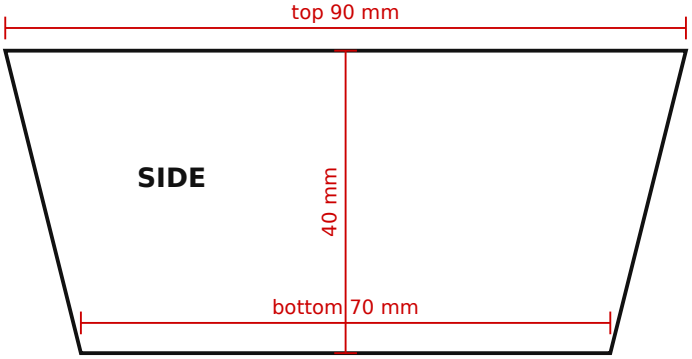


Axle hanger 1

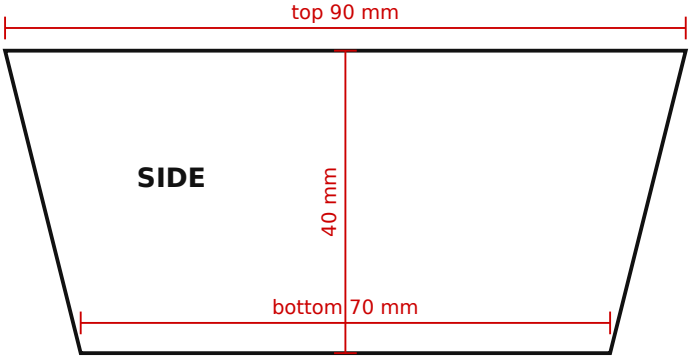


Axle hanger 2

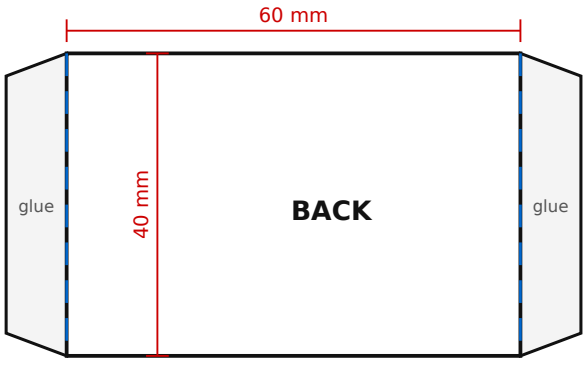
hole 8 mm up from bottom edge



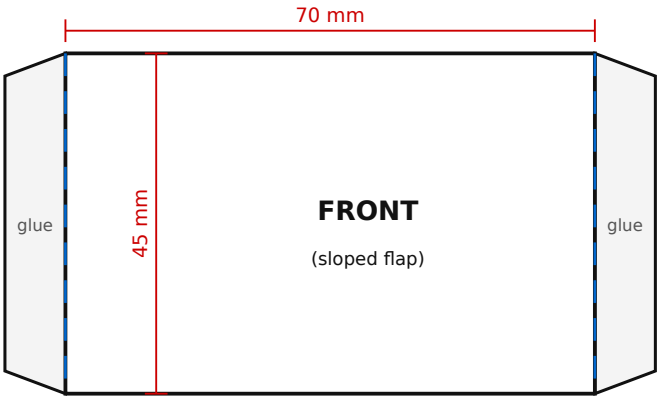
Side wall 1 (trapezoid)



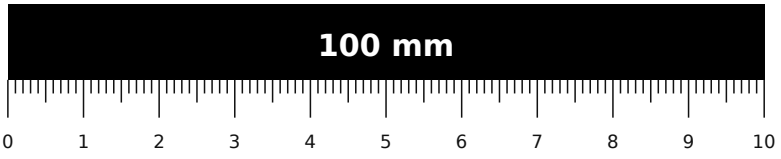
Side wall 2 (trapezoid)



Back wall 60x40 + tabs



Front flap 70x45 + tabs



20 mm

Calibration: the black bar must measure exactly 100 mm and the square 20x20 mm.
If not, reprint at 100% (Actual size).
Use card ~1 mm (cereal box) for parts.

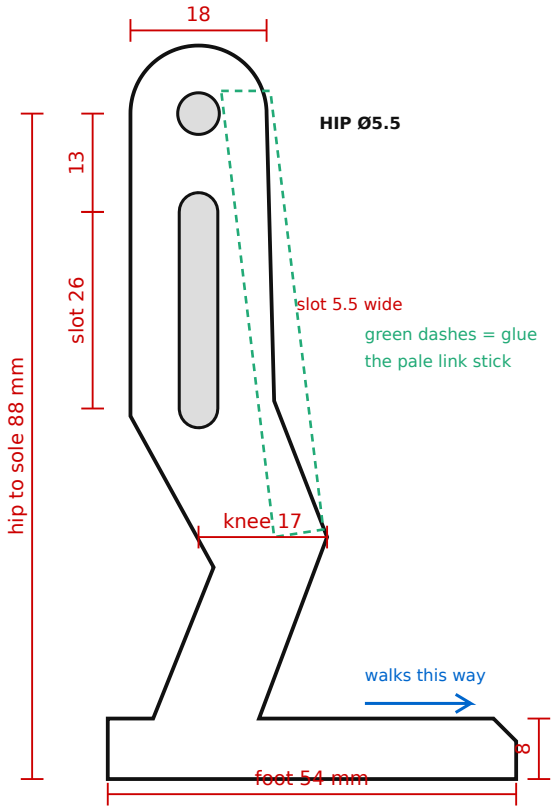
Robotics Project #13 · Templates 2/5 (1:1 scale)

Wheelbarrow Walker Robot

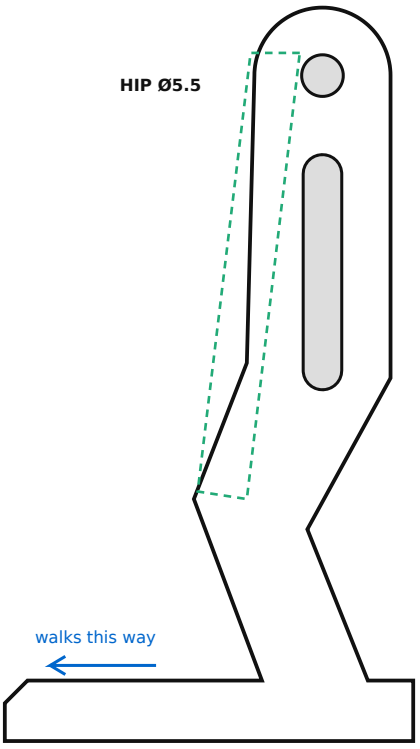
PRINT AT 100% / ACTUAL SIZE
Do NOT 'fit to page'
Check the 100 mm ruler first!

cut (solid) — fold (dashed) — hole: grown-up pokes with a pencil/awl — red = size in mm

3 mm corrugated cardboard. Trace and cut each leg TWICE, glue the two layers = one strong 6 mm leg.



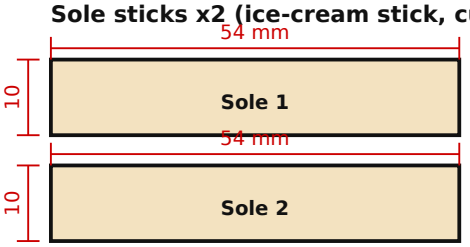
LEFT leg (cut 2)



RIGHT leg (cut 2, mirror)

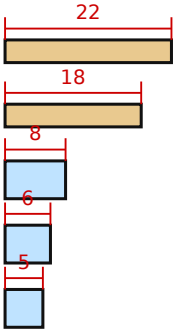
Hip hole, slot and sole must stay on one straight line (the centre line) - that is what makes the foot glide flat.

Sole sticks x2 (ice-cream stick, cut)



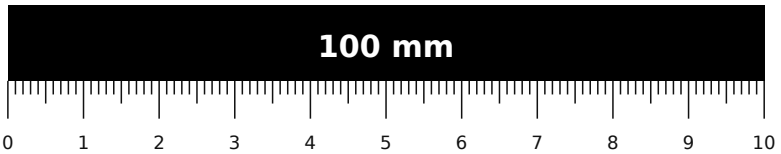
Glue a sole stick under each long foot (toe forward).
GRIP: 2 rubber bands round each foot -
the swinging foot pushes, so it must not slide.

Skewer Ø3 + straw Ø5 - true length:



- **Hip peg (skewer) x2**
- **Crank pin (skewer) x2**
- **Hip sleeve (straw) x2**
- **Roller (straw) x2**
- **Spacer (straw) x2**

Wheel axle: skewer 110 mm (as before). Stoppers: 4 cardboard buttons Ø10 (sheet 3) glued on pin/peg ends.



20 mm

Calibration: the black bar must measure exactly 100 mm and the square 20x20 mm.
If not, reprint at 100% (Actual size).
Use card ~1 mm (cereal box) for parts.

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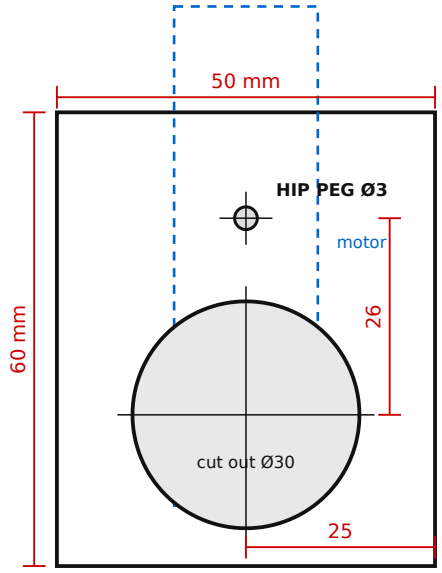
Robotics Project #13 · Templates 3/5 (1:1 scale)

Wheelbarrow Walker Robot

PRINT AT 100% / ACTUAL SIZE
Do NOT 'fit to page'
Check the 100 mm ruler first!

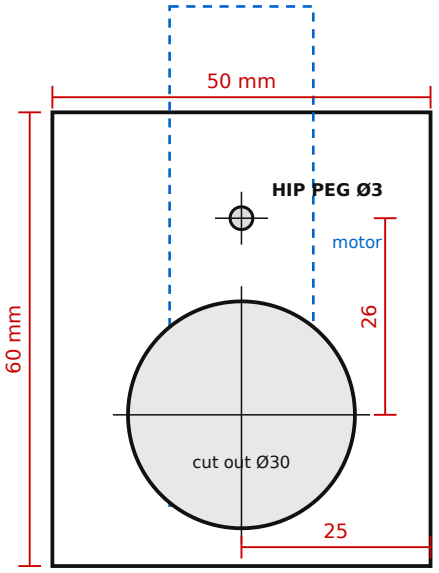
cut (solid) — — — fold (dashed) ○ hole: grown-up pokes with a pencil/awl red = size in mm

3 mm cardboard. Hip plates glue on the two shaft faces of the TT motor (gearbox end DOWN).



Hip plate LEFT

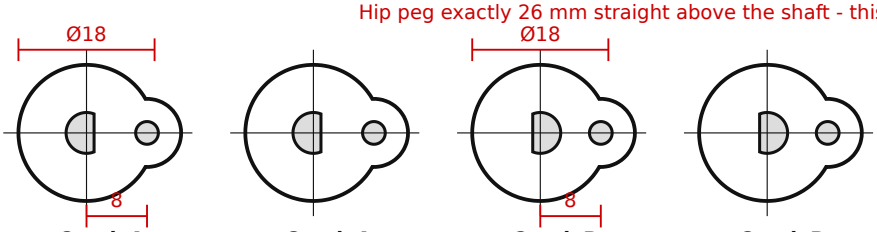
Ø30 round the shaft; blue = motor (66 long)



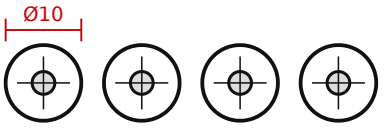
Hip plate RIGHT

Ø30 round the shaft; blue = motor (66 long)

Hip peg exactly 26 mm straight above the shaft - this sets the swing.



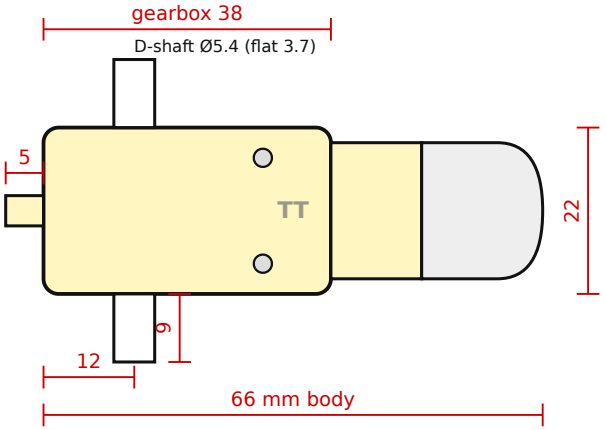
Crank A **Crank A** **Crank B** **Crank B**



Button stoppers x4 (Ø3 hole)

Glue A+A and B+B = two 6 mm cranks. D-hole Ø5.4, 3.7 across the flat: snug on the shaft + hot glue.

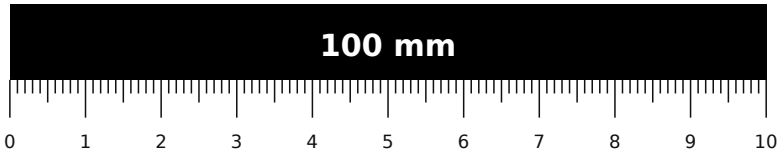
A has the flat on the lobe side, B away from it: same shaft = pins OPPOSITE = 180° apart.



CHECK YOUR MOTOR

Lay it on the drawing: body, both white shafts and the rear nub should match (± 1 mm).

Different? Move the hip peg so it is still 30 mm above the shaft.



20 mm

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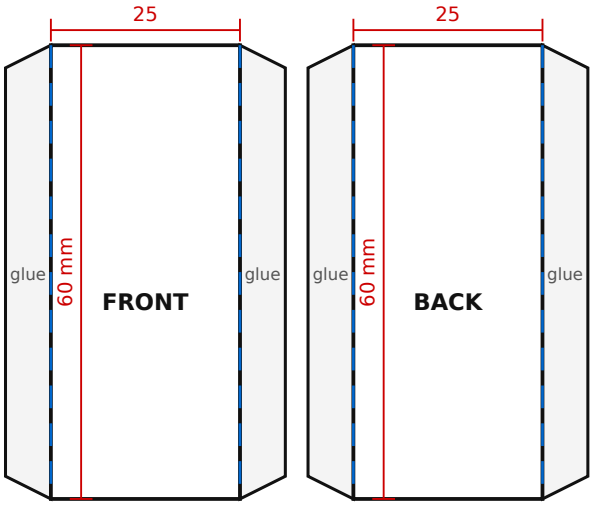
Robotics Project #13 · Templates 4/5 (1:1 scale)

Wheelbarrow Walker Robot

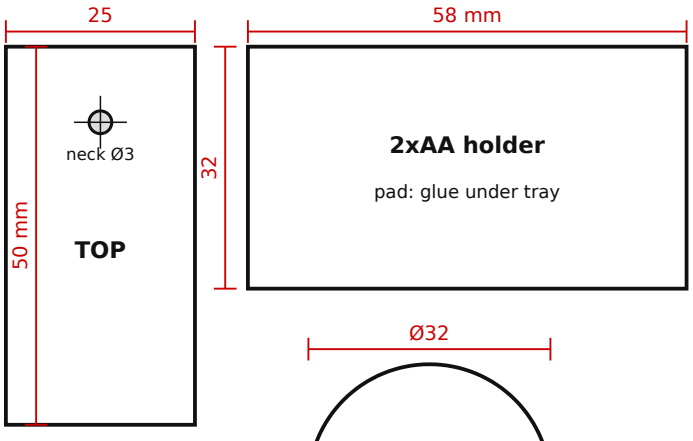
PRINT AT 100% / ACTUAL SIZE
Do NOT 'fit to page'
Check the 100 mm ruler first!

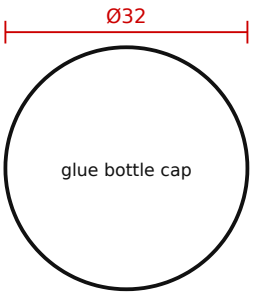
cut (solid) — fold (dashed) — hole: grown-up pokes with a pencil/awl — red = size in mm

Torso covers, battery pad, hat - and a 1:1 side view to check your leg before gluing.

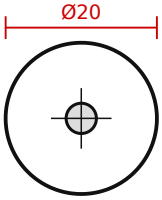


Torso covers 25x60 (+6 mm tabs glue to plate edges)

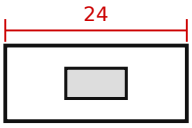




Hat brim



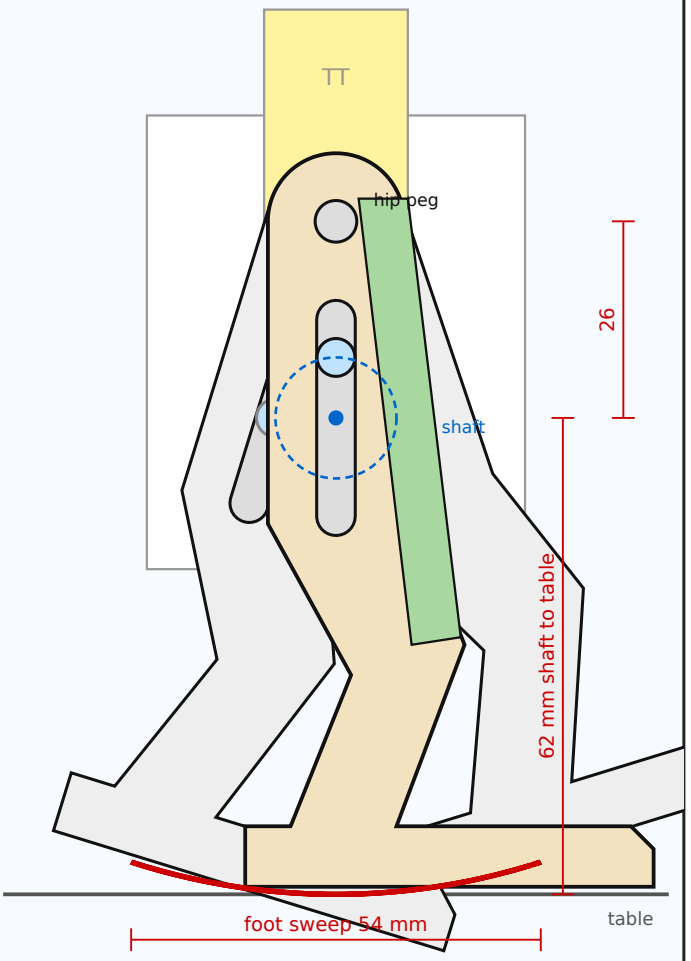
Wheel washer x2 (Ø4)



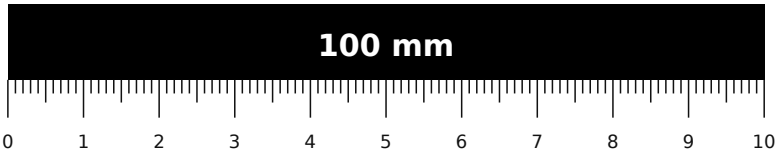
Switch plate (cut 8x4 slot)

1:1 SWING CHECK (side view)

lay your leg on this drawing



Pale legs = crank pin at front / back.
Red = foot path: 54 mm swing, only 4 mm rise.
Walks backwards? Swap the 2 motor wires.



20 mm

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If not, reprint at 100% (Actual size).
Use card ~1 mm (cereal box) for parts.

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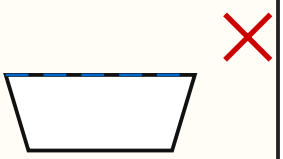
Robotics Project #13 · Templates 5/5 (1:1 scale)

Wheelbarrow Walker Robot

PRINT AT 100% / ACTUAL SIZE
Do NOT 'fit to page'
Check the 100 mm ruler first!

cut (solid) — fold (dashed) — hole: grown-up pokes with a pencil/awl — red = size in mm

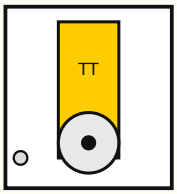
1 Cut and fold the barrow tray (sheet 1). Glue the tabs inside.



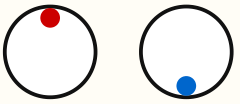
2 Skewer axle through a straw under the tray; glue a wheel on each end. Wheels roll freely.



3 Hot-glue a hip plate (sheet 3) on each flat side of the TT motor, motor upright, shaft in the Ø30 hole.



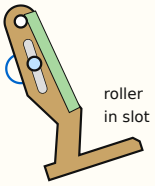
4 Glue crank A on one shaft end, crank B on the other: the pins must point opposite ways.



5 Glue a 22 mm hip peg in each hip hole. Add a 5 mm straw spacer, then the leg (hip sleeve in its Ø5.5 hole).



6 Crank pin + straw roller sits in the leg slot. Glue button stoppers on peg and pin ends.



7 Glue torso covers, arms at the shoulders, hands onto the handles with both feet on the table.

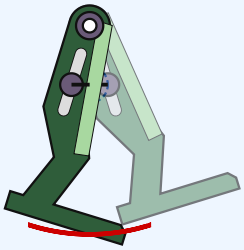


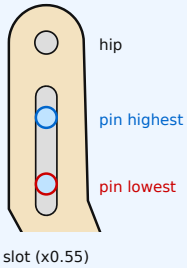
8 2xAA + switch under the tray, wires along a handle to the motor. Switch on and he walks!



ON = walk!

How the legs walk





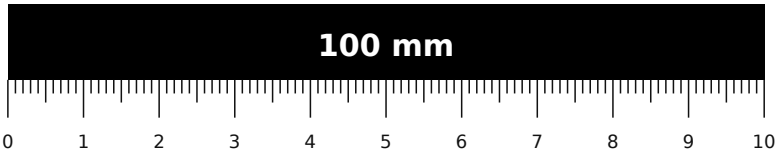
The motor turns both cranks, half a turn apart. The crank pin (with a straw roller) slides up and down in the leg slot and pushes the leg front and back. The leg swings like a pendulum from the hip peg: the foot (red) sweeps 54 mm along the table and rises only 4 mm. One foot pushes back while the other swings forward - and his hands push the barrow! Roller never leaves the slot.

Parts checklist

- ☐ 3 mm cardboard
- ☐ 3 ice-cream sticks
- ☐ 2 wheels (caps/65 mm)
- ☐ Skewers 3 mm
- ☐ Paper straw
- ☐ TT motor (double shaft)
- ☐ 2xAA holder + batteries
- ☐ Slide switch + wire
- ☐ Rubber bands (feet)
- ☐ Ping-pong ball + cap
- ☐ Hot glue (grown-up)
- ☐ Pencil, scissors

Safety first!

- A grown-up uses the hot glue gun, craft knife, and pokes the holes.
- Snip the sharp skewer tips off.
- Fingers away from the cranks when ON.
- Batteries the right way round; never join the wires; take them out after.
- Keep small parts from little kids.



20 mm

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